

GraphML Spatial Analysis Report

Howard Floor Plan

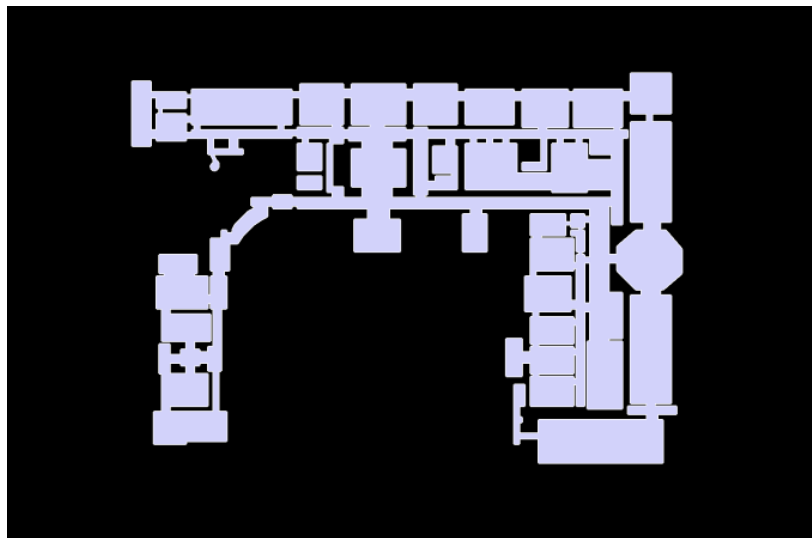
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This study explores the spatial network properties of the Howard floor plan using GraphML analysis methods. The aim of the exercise was to understand how circulation, accessibility, and spatial importance can be interpreted through graph-based analysis rather than conventional architectural drawings alone.

The floor plan was converted into a graph structure where spaces and circulation points act as nodes, while connections between them act as edges. Three analytical methods were applied:

- Shortest Path Analysis
- Degree Centrality Analysis
- Betweenness Centrality Analysis

These analyses help reveal how movement flows through the building and which spaces play critical roles in circulation and accessibility.

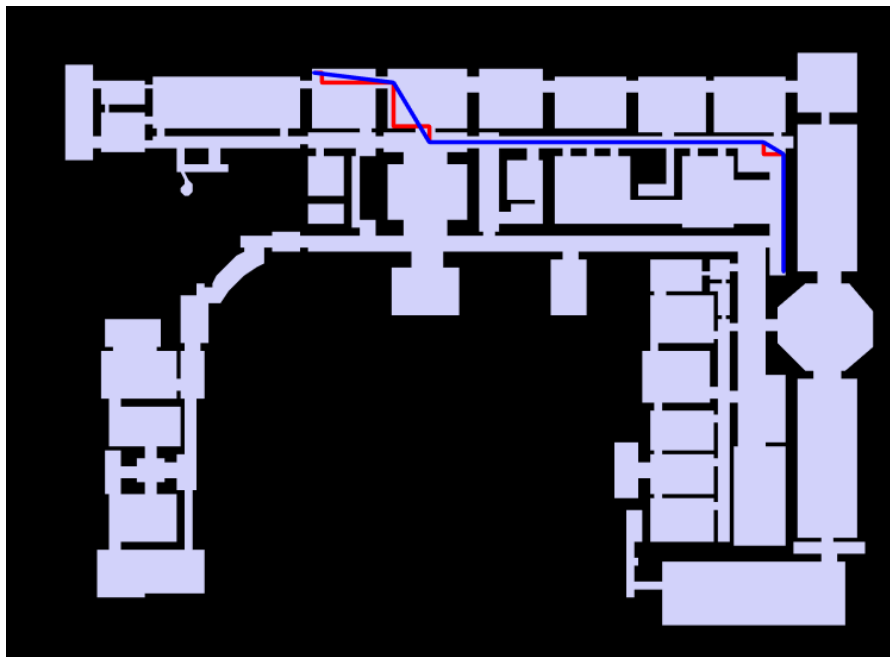


1. Shortest Path Analysis

The shortest path analysis was used to identify the most efficient route between selected nodes within the Howard floor plan. The red and blue lines illustrate the calculated circulation path across the upper corridor system.

Observations

- The analysis demonstrates that the upper horizontal corridor acts as the primary circulation spine of the building.
- Most efficient movement routes rely heavily on this central corridor because it provides the least resistance and shortest travel distance between spaces.
- Directional changes occur mainly at corridor intersections and node transitions, highlighting key turning points within the network.
- The right vertical corridor acts as a major connector linking multiple spatial clusters together.



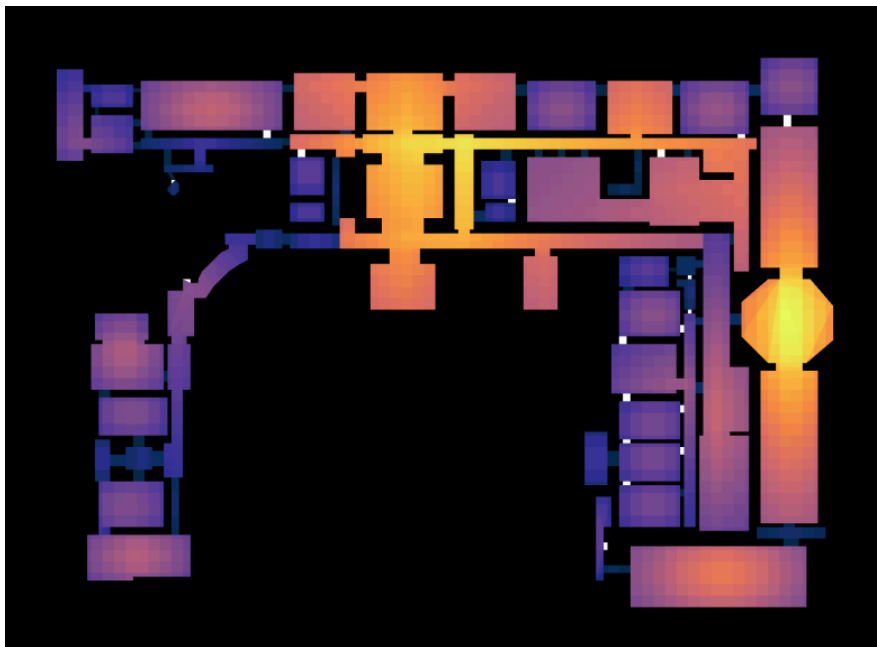
2. Degree Centrality Analysis

The degree centrality analysis measures how many direct connections each node has within the network. Higher degree values indicate spaces with greater immediate accessibility and stronger local connectivity.

In the visualisation, brighter and warmer colours represent higher centrality values.

Observations

- The central upper intersection area displays the highest degree centrality.
- Several junction nodes connecting corridors and room clusters show elevated connectivity values.
- Peripheral rooms and isolated corridor ends show much lower centrality.



3. Betweenness Centrality Analysis

The dark navy visualisation represents betweenness centrality analysis. This method measures how often a node lies on the shortest paths between other nodes in the network.

Nodes with high betweenness act as bridges or control points within circulation systems.

Observations

- The major corridor intersections contain the highest betweenness values.
- The long horizontal corridor and right-side vertical circulation spine dominate the network flow.
- Several nodes act as bottlenecks where movement must pass through before reaching other areas.
- Peripheral rooms contribute very little to network mediation.

